

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

Let $\vec{a} = -5\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 4\hat{k}$ and $\vec{c} = (((\vec{a} \times \vec{b}) \times \hat{i}) \times \hat{i}) \times \hat{i}$. Then $\vec{c} \cdot (-\hat{i} + \hat{j} + \hat{k})$ is equal to

(1) -12

(2) -10

(3) -13

(4) -15

Q2 - 2024 (01 Feb Shift 2)

Consider a $\triangle ABC$ where $A(1, 2, 3)$, $B(-2, 8, 0)$ and $C(3, 6, 7)$. If the angle bisector of $\angle BAC$ meets the line BC at D , then the length of the projection of the vector $\vec{AD}$ on the vector $\vec{AC}$ is:

(1) $\frac{37}{2\sqrt{38}}$

(2) $\frac{\sqrt{38}}{2}$

(3) $\frac{39}{2\sqrt{38}}$

(4) $\sqrt{19}$

Q3 - 2024 (01 Feb Shift 2)

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ and $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three vectors such that $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$. If the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j} + \hat{k}$ is θ , then the greatest integer less than or equal to $\tan^2 \theta$ is :

Q4 - 2024 (27 Jan Shift 1)

Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3(\hat{i} - \hat{j} + \hat{k})$. Let $\vec{c}$ be the vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$. Then $\vec{a} \cdot ((\vec{c} \times \vec{b}) - \vec{b} - \vec{c})$ is equal to :

(1) 32

(2) 24

(3) 20

(4) 36

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Q5 - 2024 (27 Jan Shift 1)

The least positive integral value of α , for which the angle between the vectors $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ and $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ is acute, is _____

Q6 - 2024 (29 Jan Shift 1)

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear if $\vec{a} + 5\vec{b}$ is collinear with $\vec{c}$, $\vec{b} + 6\vec{c}$ is collinear with $\vec{a}$ and $\vec{a} + \alpha\vec{b} + \beta\vec{c} = \vec{0}$, then $\alpha + \beta$ is equal to

(1) 35

(2) 30

(3) -30

(4) -25

Q7 - 2024 (29 Jan Shift 2)

Let $\vec{OA} = \vec{a}$, $\vec{OB} = 12\vec{a} + 4\vec{b}$ and $\vec{OC} = \vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then

$\frac{\text{area of the quadrilateral OABC}}{\text{area of S}}$ is equal to _____

(1) 6

(2) 10

(3) 7

(4) 8

Q8 - 2024 (29 Jan Shift 2)

Let a unit vector $\hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$ make angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$, then $|\hat{u} - \vec{v}|^2$ is equal to

(1) $\frac{11}{2}$ (2) $\frac{5}{2}$

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(3) 9

(4) 7

Q9 - 2024 (30 Jan Shift 1)

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ be two vectors such that $|\vec{a}| = 1$; $\vec{a} \cdot \vec{b} = 2$ and $|\vec{b}| = 4$. If $\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to :

(1) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(2) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

(3) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$

(4) $\cos^{-1}\left(\frac{2}{3}\right)$

Q10 - 2024 (30 Jan Shift 1)

Let $A(2, 3, 5)$ and $C(-3, 4, -2)$ be opposite vertices of a parallelogram ABCD if the diagonal $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$ then the area of the parallelogram is equal to

(1) $\frac{1}{2}\sqrt{410}$

(2) $\frac{1}{2}\sqrt{474}$

(3) $\frac{1}{2}\sqrt{586}$

(4) $\frac{1}{2}\sqrt{306}$

Q11 - 2024 (30 Jan Shift 2)

Let $\vec{a} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$, $\alpha, \beta \in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $|\vec{b}|^2 = 6$. If $\vec{a} \cdot \vec{b} = 3\sqrt{2}$, then the value of $(\alpha^2 + \beta^2) |\vec{a} \times \vec{b}|^2$ is equal to

(1) 90

(2) 75

(3) 95

(4) 85

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Q12 - 2024 (30 Jan Shift 2)

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{b}| = 1$ and $|\vec{b} \times \vec{a}| = 2$. Then $|(\vec{b} \times \vec{a}) - \vec{b}|^2$ is equal to

(1)

(2)

(3)

(4)

(1) 3

(2) 5

(3) 1

(4) 4

Q13 - 2024 (31 Jan Shift 1)

Let $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = 4\hat{i} + \hat{j} + 7\hat{k}$ and $\vec{c} = \hat{i} - 3\hat{j} + 4\hat{k}$ be three vectors. If a vector $\vec{p}$ satisfies $\vec{p} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{p} \cdot \vec{a} = 0$, then $\vec{p} \cdot (\hat{i} - \hat{j} - \hat{k})$ is equal to

(1) 24

(2) 36

(3) 28

(4) 32

Q14 - 2024 (31 Jan Shift 1)

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α , then $192 \sin^2 \alpha$ is equal to _____.

Q15 - 2024 (31 Jan Shift 2)

Let $\vec{a} = 3\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$ and $(\vec{a} - \vec{b} + \hat{i}) \cdot \vec{c} = -3$. Then $|\vec{c}|^2$ is equal to _____.

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Answer Key

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Q1 (1) mathongo /// ma**Q2** (1) o /// mathongo **Q3** (38) mathongo /// mat**Q4** (2) /// mathongo

Q5 (5) athongo /// ma**Q6** (1) o /// mathongo **Q7** (4) athongo /// mat**Q8** (2) /// mathongo

Q9 (3) athongo /// ma**Q10** (2) /// mathongo **Q11** (1) athongo /// mat**Q12** (2) /// mathongo

Q13 (4) thongo /// ma**Q14** (48) /// mathongo **Q15** (38) thongo /// mathongo /// mathongo

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Solutions

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Q1

$$\vec{a} = -5\hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{b} = \hat{i} + 2\hat{j} - 4\hat{k}$$

$$(\vec{a} \times \vec{b}) \times \hat{i} = (\vec{a} \cdot \hat{i})\vec{b} - (\vec{b} \cdot \hat{i})\vec{a}$$

$$= -5\vec{b} - \vec{a}$$

$$= (((-5\vec{b} - \vec{a}) \times \hat{i}) \times \hat{i})$$

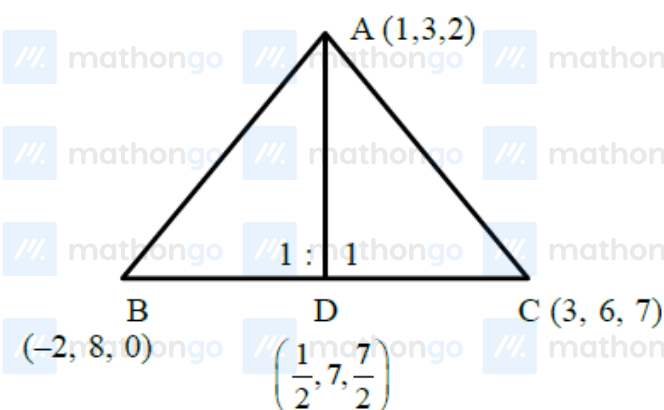
$$= ((-11\hat{j} + 23\hat{k}) \times \hat{i}) \times \hat{i}$$

$$\Rightarrow (11\hat{k} + 23\hat{j}) \times \hat{i}$$

$$\Rightarrow (11\hat{j} - 23\hat{k})$$

$$\vec{c} \cdot (-\hat{i} + \hat{j} + \hat{k}) = 11 - 23 = -12$$

Q2



$$A(1, 3, 2); B(-2, 8, 0); C(3, 6, 7);$$

$$\vec{AC} = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

$$AB = \sqrt{9 + 25 + 4} = \sqrt{38}$$

$$AC = \sqrt{4 + 9 + 25} = \sqrt{38}$$

$$\vec{AD} = \frac{1}{2}\hat{i} - 4\hat{j} - \frac{3}{2}\hat{k} = \frac{1}{2}(\hat{i} - 8\hat{j} - 3\hat{k})$$

Length of projection of $\vec{AD}$ on $\vec{AC}$

$$= \left| \frac{\vec{AD} \cdot \vec{AC}}{|\vec{AC}|} \right| = \frac{37}{2\sqrt{38}}$$

Q3

Solutions

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$$\vec{a} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{b} = \hat{i} + 8\hat{j} + 2\hat{k}$$

$$\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$$

$$\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$$

$$(\vec{b} - \vec{c}) \times \vec{a} = 0$$

$$\vec{b} - \vec{c} = \lambda \vec{a}$$

$$\vec{b} = \vec{c} + \lambda \vec{a}$$

$$-\hat{i} - 8\hat{j} + 2\hat{k} = (4\hat{i} + c_2\hat{j} + c_3\hat{k}) + \lambda(\hat{i} + \hat{j} + \hat{k})$$

$$\lambda + 4 = -1 \Rightarrow \lambda = -5$$

$$\lambda + c_2 = -8 \Rightarrow c_2 = -3$$

$$\lambda + c_3 = 2 \Rightarrow c_3 = 7$$

$$\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$$

$$\cos \theta = \frac{12 - 12 + 7}{\sqrt{26} \cdot \sqrt{74}} = \frac{7}{\sqrt{26} \cdot \sqrt{74}} = \frac{7}{2\sqrt{481}}$$

$$\tan^2 \theta = \frac{625 \times 3}{49}$$

$$[\tan^2 \theta] = 38$$

Q4

$$\vec{a} \cdot [(\vec{c} \times \vec{b}) - \vec{b} - \vec{c}]$$

$$\vec{a} \cdot (\vec{c} \times \vec{b}) - \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} \dots (i)$$

$$\text{given } \vec{a} \times \vec{c} = \vec{b}$$

$$\Rightarrow (\vec{a} \times \vec{c}) \cdot \vec{b} = \vec{b} \cdot \vec{b} = |\vec{b}|^2 = 27$$

$$\Rightarrow \vec{a} \cdot (\vec{c} \times \vec{b}) = [\vec{a} \quad \vec{c} \quad \vec{b}] = (\vec{a} \times \vec{c}) \cdot \vec{b} = 27 \dots (ii)$$

$$\text{Now } \vec{a} \cdot \vec{b} = 3 - 6 + 3 = 0 \dots (iii)$$

$$\vec{a} \cdot \vec{c} = 3 \dots (iv) \text{ (given)}$$

$$\text{By (i), (ii), (iii) \& (iv)}$$

$$27 - 0 - 3 = 24$$

Q5

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Solutions

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$$\cos \theta = \frac{(\alpha \hat{i} - 2\hat{j} + 2\hat{k}) \cdot (\alpha \hat{i} + 2\alpha \hat{j} - 2\hat{k})}{\sqrt{\alpha^2 + 4 + 4}\sqrt{\alpha^2 + 4\alpha^2 + 4}}$$

$$\cos \theta = \frac{\alpha^2 - 4\alpha - 4}{\sqrt{\alpha^2 + 8}\sqrt{5\alpha^2 + 4}}$$

$$\Rightarrow \alpha^2 - 4\alpha - 4 > 0$$

$$\Rightarrow \alpha^2 - 4\alpha + 4 > 8 \Rightarrow (\alpha - 2)^2 > 8$$

$$\Rightarrow \alpha - 2 > 2\sqrt{2} \text{ or } \alpha - 2 < -2\sqrt{2}$$

$$\alpha > 2 + 2\sqrt{2} \text{ or } \alpha < 2 - 2\sqrt{2}$$

$$\alpha \in (-\infty, -0.82) \cup (4.82, \infty)$$

Least positive integral value of $\alpha \Rightarrow 5$

Q6

$$\vec{a} + 5\vec{b} = \lambda \vec{c}$$

$$\vec{b} + 6\vec{c} = \mu \vec{a}$$

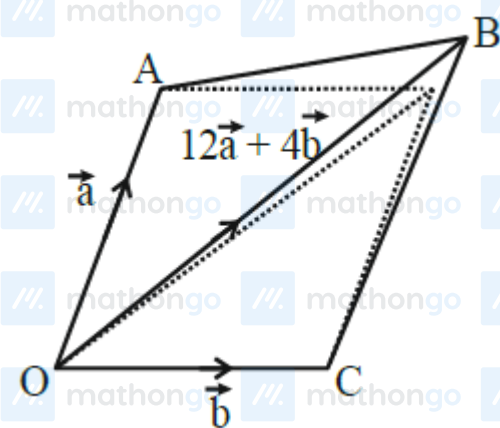
Eliminating $\vec{a}$

$$\lambda \vec{c} - 5\vec{b} = \frac{6}{\mu} \vec{c} + \frac{1}{\mu} \vec{b}$$

$$\therefore \mu = \frac{-1}{5}, \lambda = -30$$

$$\alpha = 5, \beta = 30$$

Q7



Area of parallelogram, $S = |\vec{a} \times \vec{b}|$

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Solutions

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$$\text{Area of quadrilateral} = \text{Area}(\triangle OAB) + \text{Area}(\triangle OBC)$$

$$= \frac{1}{2} \{ |\vec{a} \times (12\vec{a} + 4\vec{b})| + |\vec{b} \times (12\vec{a} + 4\vec{b})| \}$$

$$= 8 |(\vec{a} \times \vec{b})|$$

$$\text{Ratio} = \frac{8 |(\vec{a} \times \vec{b})|}{|(\vec{a} \times \vec{b})|} = 8$$

Q8

$$\text{Unit vector } \hat{u} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{p}_1 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}, \vec{p}_2 = \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$$

$$\vec{p}_3 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

$$\text{Now angle between } \hat{u} \text{ and } \vec{p}_1 = \frac{\pi}{2}$$

$$\hat{u} \cdot \vec{p}_1 = 0 \Rightarrow \frac{x}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0$$

$$\Rightarrow x + z = 0 \dots (i)$$

$$\text{Angle between } \hat{u} \text{ and } \vec{p}_2 = \frac{\pi}{3}$$

$$\hat{u} \cdot \vec{p}_2 = |\hat{u}| \cdot |\vec{p}_2| \cos \frac{\pi}{3}$$

$$\Rightarrow \frac{y}{\sqrt{2}} + \frac{z}{\sqrt{2}} = \frac{1}{2} \Rightarrow y + z = \frac{1}{\sqrt{2}} \dots (ii)$$

$$\text{Angle between } \hat{u} \text{ and } \vec{p}_3 = \frac{2\pi}{3}$$

$$\hat{u} \cdot \vec{p}_3 = |\hat{u}| \cdot |\vec{p}_3| \cos \frac{2\pi}{3}$$

$$\Rightarrow \frac{x}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{-1}{2} \Rightarrow x + y = \frac{-1}{\sqrt{2}} \dots (iii)$$

from equation (i), (ii) and (iii) we get

$$x = \frac{-1}{\sqrt{2}} \quad y = 0 \quad z = \frac{1}{\sqrt{2}}$$

$$\text{Thus } \hat{u} - \vec{v} = \frac{-1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k} - \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$$

$$\hat{u} - \vec{v} = \frac{-2}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$$

$$\therefore |\hat{u} - \vec{v}|^2 = \left(\sqrt{\frac{4}{2} + \frac{1}{2}} \right)^2 = \frac{5}{2}$$

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Solutions

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Q9

$$\text{Given } |\vec{a}| = 1, |\vec{b}| = 4, \vec{a} \cdot \vec{b} = 2$$

$$\vec{c} = 2(\vec{a} \times \vec{b}) - 3\vec{b}$$

Dot product with $\vec{a}$ on both sides

$$\vec{c} \cdot \vec{a} = -6 \dots (1)$$

Dot product with $\vec{b}$ on both sides

$$\vec{b} \cdot \vec{c} = -48 \dots (2)$$

$$\vec{c} \cdot \vec{c} = 4|\vec{a} \times \vec{b}|^2 + 9|\vec{b}|^2$$

$$|\vec{c}|^2 = 4[|\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2] + 9|\vec{b}|^2$$

$$|\vec{c}|^2 = 4[(1)(4)^2 - (2)^2] + 9(16)$$

$$|\vec{c}|^2 = 4[12] + 144$$

$$|\vec{c}|^2 = 48 + 144$$

$$|\vec{c}|^2 = 192$$

$$\therefore \cos \theta = \frac{\vec{b} \cdot \vec{c}}{|\vec{b}| |\vec{c}|}$$

$$\therefore \cos \theta = \frac{-48}{\sqrt{192} \cdot 4}$$

$$\therefore \cos \theta = \frac{-48}{8\sqrt{3} \cdot 4}$$

$$\therefore \cos \theta = \frac{-3}{2\sqrt{3}}$$

$$\therefore \cos \theta = \frac{-\sqrt{3}}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$$

Q10

$$\text{Area} = |\vec{AC} \times \vec{BD}|$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -1 & 7 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= \frac{1}{2} |-17\hat{i} - 8\hat{j} + 11\hat{k}| = \frac{1}{2} \sqrt{474}$$

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Solutions

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Q11

$$|\vec{b}|^2 = 6; |\vec{a}| |\vec{b}| \cos \theta = 3\sqrt{2}$$

$$|\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta = 18$$

$$|\vec{a}|^2 = 6$$

$$\text{Also } 1 + \alpha^2 + \beta^2 = 6$$

$$\alpha^2 + \beta^2 = 5$$

to find

$$(\alpha^2 + \beta^2) |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta$$

$$= (5)(6)(6) \left(\frac{1}{2}\right)$$

$$= 90$$

Q12

$$|\vec{b}| = 1 \& |\vec{b} \times \vec{a}| = 2$$

$$(\vec{b} \times \vec{a}) \cdot \vec{b} = \vec{b} \cdot (\vec{b} \times \vec{a}) = 0$$

$$|(\vec{b} \times \vec{a}) - \vec{b}|^2 = |\vec{b} \times \vec{a}|^2 + |\vec{b}|^2$$

$$= 4 + 1 = 5$$

Q13

$$\vec{p} \times \vec{b} - \vec{c} \times \vec{b} = \vec{0}$$

$$(\vec{p} - \vec{c}) \times \vec{b} = \vec{0}$$

$$\vec{p} - \vec{c} = \lambda \vec{b} \Rightarrow \vec{p} = \vec{c} + \lambda \vec{b}$$

$$\text{Now, } \vec{p} \cdot \vec{a} = 0 \text{ (given)}$$

$$\text{So, } \vec{c} \cdot \vec{a} + \lambda \vec{a} \cdot \vec{b} = 0$$

$$(3 - 3 - 8) + \lambda(12 + 1 - 14) = 0$$

$$\lambda = -8$$

$$\vec{p} = \vec{c} - 8\vec{b}$$

$$\vec{p} = -31\hat{i} - 11\hat{j} - 52\hat{k}$$

$$\text{So, } \vec{p} \cdot (\hat{i} - \hat{j} - \hat{k})$$

$$= -31 + 11 + 52$$

$$= 32$$

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Q14

$$\vec{b} \cdot \vec{c} = (2\vec{a} \times \vec{b}) \cdot \vec{b} - 3|\vec{b}|^2$$

$$|\vec{b}||\vec{c}| \cos \alpha = -3|\vec{b}|^2$$

$$|\vec{c}| \cos \alpha = -12, \text{ as } |\vec{b}| = 4$$

$$\vec{a} \cdot \vec{b} = 2$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$|\vec{c}|^2 = |(2\vec{a} \times \vec{b}) - 3\vec{b}|^2$$

$$= 64 \times \frac{3}{4} + 144 = 192$$

$$|\vec{c}|^2 \cos^2 \alpha = 144$$

$$192 \cos^2 \alpha = 144$$

$$192 \sin^2 \alpha = 48$$

Q15

$$(\vec{a} + \vec{b}) \times \vec{c} = 2(\vec{a} \times \vec{b}) + 24\hat{j} - 6\hat{k}$$

$$(5\hat{i} + \hat{j} + 4\hat{k}) \times \vec{c} = 2(7\hat{i} - 7\hat{j} - 7\hat{k}) + 24\hat{j} - 6\hat{k}$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 1 & 4 \\ x & y & z \end{vmatrix} = 14\hat{i} + 10\hat{j} - 20\hat{k}$$

$$\Rightarrow \hat{i}(z - 4y) - \hat{j}(5z - 4x) + \hat{k}(5y - x) = 14\hat{i} + 10\hat{j} - 20\hat{k}$$

$$z - 4y = 14, 4x - 5z = 10, 5y - x = -20$$

$$(a - b + i) \cdot \vec{c} = -3$$

$$(2\hat{i} + 3\hat{j} - 2\hat{k}) \cdot \vec{c} = -3$$

$$2x + 3y - 2z = -3$$

$$\therefore x = 5, y = -3, z = 2$$

$$|\vec{c}|^2 = 25 + 9 + 4 = 38$$

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